

Similarity Search Improvement Doc

Target filter

This filter during the similarity search will ensure that the same user doesn't have their journey trivially repeated in the search results.

```
filter=f"id != {get_search_journey_id()}"
```

Similarity Scores:

Milvus returns a decimal to map the different euclidian distances.

- 22 Metres gives a distance of: 4.087632987648249e-08
- 11 metres gives a distance of: 9.837094694375992e-09
- 600 metres gives a distance of: 3.9029400795698166e-05
- 930 metres gives a distance of: 0.00013135983317624778
- 2481 metres gives a distance of: 0.0005475086509250104
- 2510 metres gives a distance of: 0.0005708029493689537

Attempts to use linear regression to find a function which can map the distance in metres onto the distance in Milvus.

- Degree 1 (Linear):
 $y = 2.344 \cdot 10^{-7}x - 4.129 \cdot 10^{-5}$
- Degree 2 (Quadratic):
 $y = 6.278 \cdot 10^{-11}x^2 + 6.925 \cdot 10^{-8}x - 3.990 \cdot 10^{-6}$
- Degree 3 (Cubic):
 $y = -7.914 \cdot 10^{-14}x^3 + 3.267 \cdot 10^{-10}x^2 - 9.886 \cdot 10^{-8}x + 1.195 \cdot 10^{-6}$
- Degree 4 (Quartic):
 $y = 1.100 \cdot 10^{-16}x^4 - 5.397 \cdot 10^{-13}x^3 + 8.677 \cdot 10^{-10}x^2 - 2.924 \cdot 10^{-7}x + 4.562 \cdot 10^{-6}$
- Degree 5 (Quintic):
 $y = 8.241 \cdot 10^{-20}x^5 - 4.274 \cdot 10^{-16}x^4 + 6.520 \cdot 10^{-13}x^3 - 1.591 \cdot 10^{-10}x^2 + 7.528 \cdot 10^{-9}x - 5.458 \cdot 10^{-8}$

This needs to be taken into consideration when returning 'similar results' depending if the final product will end up displaying them. Currently, the search works fine displaying the top similar results as are found in the database.

Winthrop House, London W12 7QB

1 MacKenzie Cl, London W12 7LZ

Add destination

Options

Send directions to your phone Copy link

via Australia Rd 1 min 259 ft

Details

Mostly flat

GPS Coordinates 1

Latitude 51.5136

Longitude -0.232

GPS Coordinates 2

Latitude 51.5138

Longitude -0.232

Calculate Distance

The distance is between the two gps coordinates is
0.02 KM or
0.01 Miles or
0.01 Nautical miles or
22.24 meters